

Standard Operating Procedure (SOP) - AFW Pump Start and Verification

SOP-ID: SOP-AFW-001 | Revision: 3 | Authority: Mandatory | Effective Date: 2025-10-01

1. Purpose

Provide a standardized method to start Auxiliary Feedwater (AFW) Pump P-101A and verify post-start parameters.

2. Scope / Applicability

Applies to AFW system in Modes 1-4. Equipment: P-101A, MOV-204A, FT-1102, PT-1102. Train: A.

3. Preconditions / Prerequisites

Item	Requirement
Alignment	Suction MOV-204A is OPEN; discharge valve in required lineup.
Power	Motor breaker BKR-4102 is CLOSED; control power available.
Instrumentation	PT-1102 and FT-1102 indicate plausible values (no instrument trouble alarm).
Communications	Operator in communication with Control Room.

4. Cautions / Warnings / Holds

CAUTION: Do not exceed discharge pressure 1500 psig. If pressure rises above limit, stop the pump and notify supervision.
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5. Procedure

5.1 Initial Conditions

Confirm AFW header pressure is 900-1100 psig and suction pressure is stable prior to start.

Step	Action / Verification
5.2.1	Verify MOV-204A position indication reads OPEN.
5.2.2	Place P-101A control switch to START. Record start time.
5.2.3	Within 10 seconds, verify motor current stabilizes below 220 A.
5.2.4	Verify FT-1102 indicates flow $\geq$ 450 gpm within 60 seconds.
5.2.5	Verify discharge pressure PT-1102 is between 1000 and 1400 psig.
5.2.6	If vibration alarm occurs ($\geq$ 5.0 mils), stop the pump and initiate troubleshooting guidance in Section 7.

6. Acceptance Criteria / Records

Record the following in the operations log: start time, stabilized current, flow, discharge pressure, and any alarms.

7. Troubleshooting Guidance (Informational)

Use this section to support diagnostic steps. It does not override mandatory operating requirements.

Symptom	Likely Causes	Checks / Tests
Low flow (< 450 gpm)	Suction restriction; valve not fully open; air in line	Check OV-204A indication; verify suction pressure; vent
High vibration (>= 5.0 mils)	Cavitation; bearing wear; misalignment	Compare suction pressure vs NPSH margin; review bearing
High current (> 220 A)	Pump binding; discharge pressure too high; check discharge	Check discharge pressure; inspect for abnormal noise; ve

Appendix A: Example Parameter Log

Timestamp	Current (A)	Flow (gpm)	Discharge Pressure (psig)	Notes
2026-02-06 09:10	205	470	1180	Normal start
2026-02-06 09:12	208	465	1175	No alarms